

DATA VISUALIZATION REPORT

Project 4 — Boardroom-Ready Insights from E-Commerce Data

DecodeLabsIndustrialTraining · Batch2026

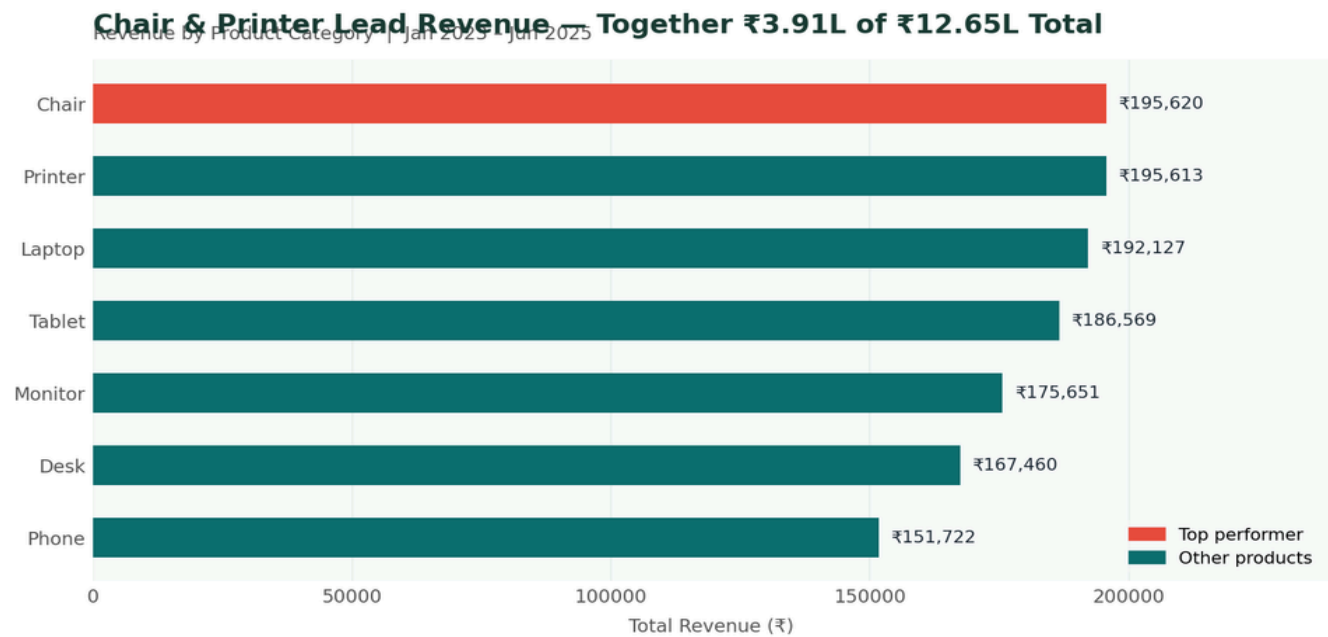
Total Revenue	Total Orders	Avg Order Value	Revenue Leakage	Top Channel
Rs12,64,762	1,200	Rs1,053.97	41.4%	Instagram

Business Context	DecodeLabs operates an e-commerce platform with 1,200 orders across 7 product categories, 5 referral channels, and 5 payment methods spanning Jan 2023 – Jun 2025. This report translates raw data into 6 visual insights to drive executive decisions on revenue growth, channel investment, and operational improvement.
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CHART SELECTION RATIONALE — WHY THESE VISUALS?

Chart	Type	Business Question Answered	Why This Chart?
1	Horizontal Bar	Which product earns the most?	Comparing values across categories — bars rank them instantly
2	Line Chart	Is revenue growing over time?	Shows trends & seasonality across 30 months clearly
3	Donut Chart	Where is revenue being lost?	Shows part-to-whole; donut centre highlights the crisis stat
4	Horizontal Bar	Which channel brings most revenue?	Side-by-side comparison with order count annotation
5	Bar Chart	Which payment method = bigger spend?	Mean values per category; reference line shows the benchmark
6	Heatmap	What drives order value most?	Correlation matrix with colour intensity = relationship strength

Chart 1 — Revenue by Product Category

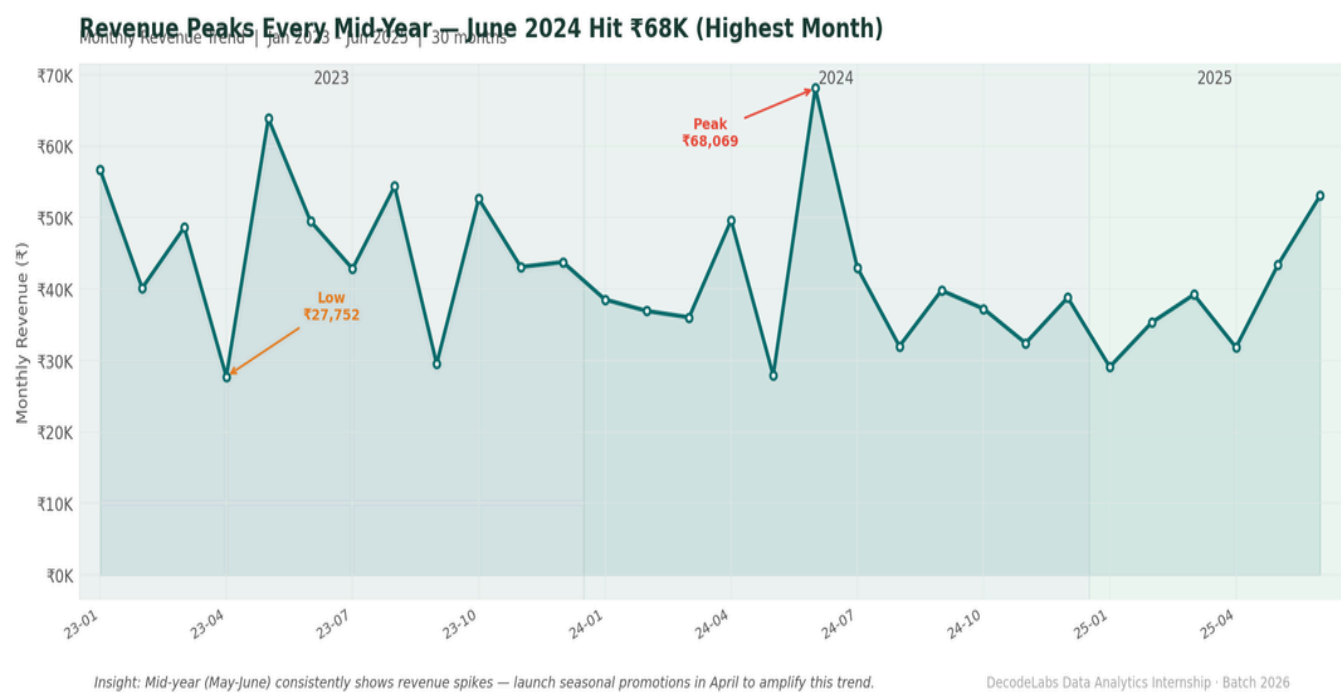


Insight: Chair & Printer each contribute ~15.5% of total revenue — prioritise inventory & bundling for these categories

Insight + Recommendation:

Chair & Printer dominate with ~Rs1.96L each (15.5% each of total revenue). Phone is the lowest performer at Rs1.52L (12%). Recommendation: Bundle Chair with Desk (complementary products), run premium campaigns for Laptop (highest avg order value).

Chart 2 — Monthly Revenue Trend (Jan 2023 – Jun 2025)



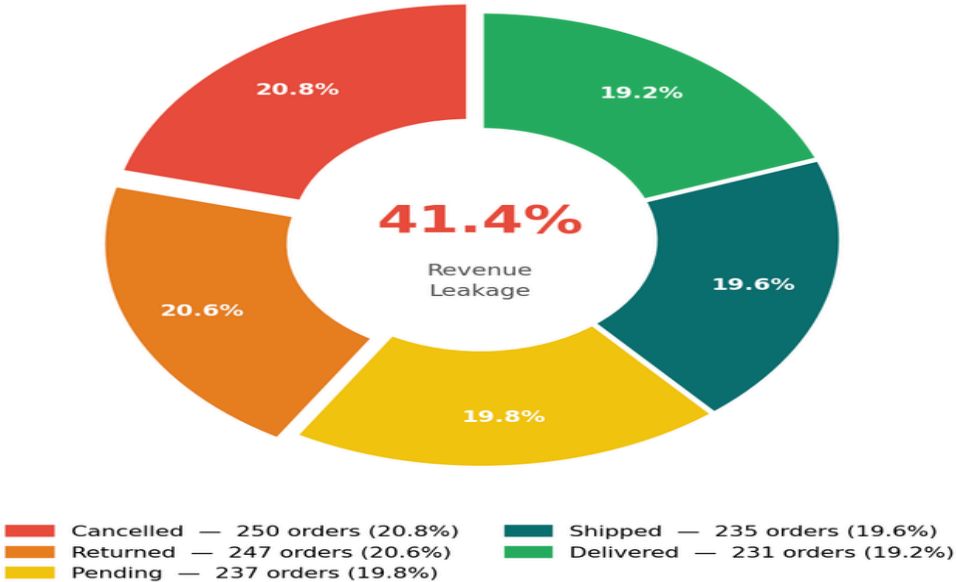
Insight +
Recommendation:

June 2024 was the peak month at Rs 68,069. A consistent mid-year spike (May-June) is visible across all 3 years. Recommendation: Launch promotional campaigns in April every year to pre-load inventory and capitalise on the mid-year demand surge.

Chart 3 — Order Status Distribution (The Revenue Crisis)

41.4% of All Orders Are Cancelled or Returned — Critical Revenue Crisis

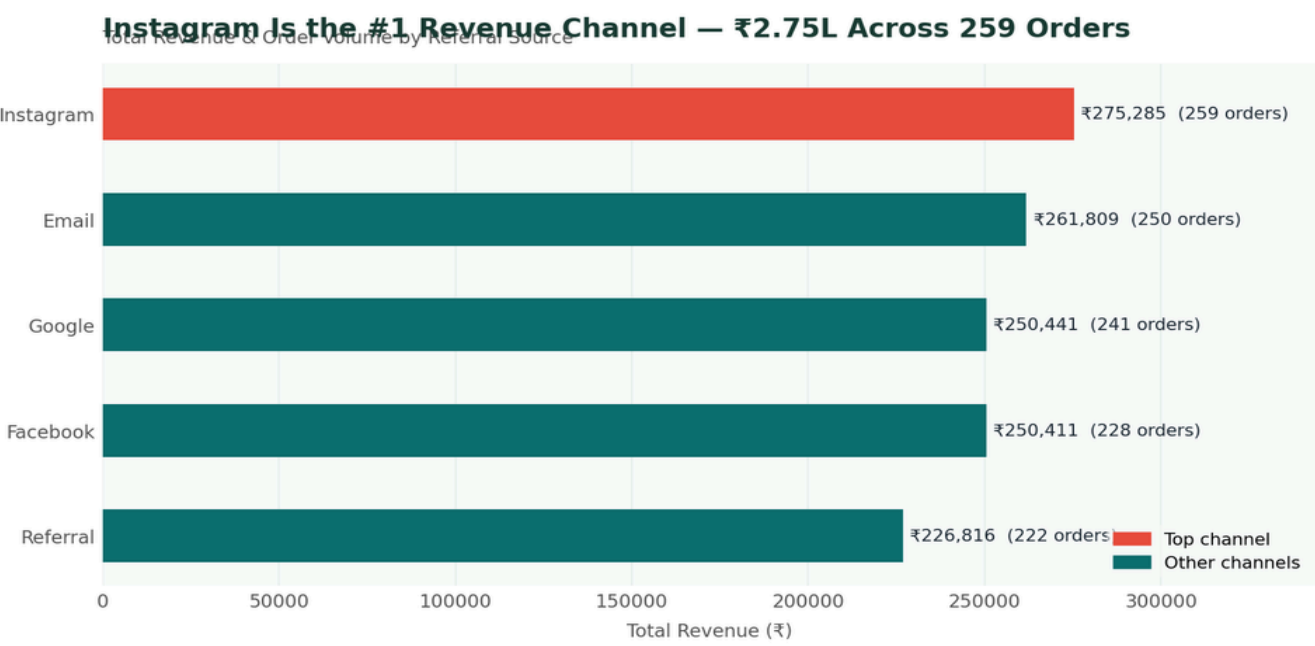
Jan 2023 – Jun 2025 | 1,200 total orders



Insight: 497 orders (41.4%) never generated revenue. Immediate audit of cancellation & return reasons required. - Batch 2026

Insight + Recommendation:	CRITICAL: 250 Cancelled + 247 Returned = 497 orders (41.4%) that generated zero net revenue. Only 231 orders (19.3%) are Delivered. Recommendation: Investigate top 3 cancellation & return reasons immediately. Add product quality checks and improved descriptions.
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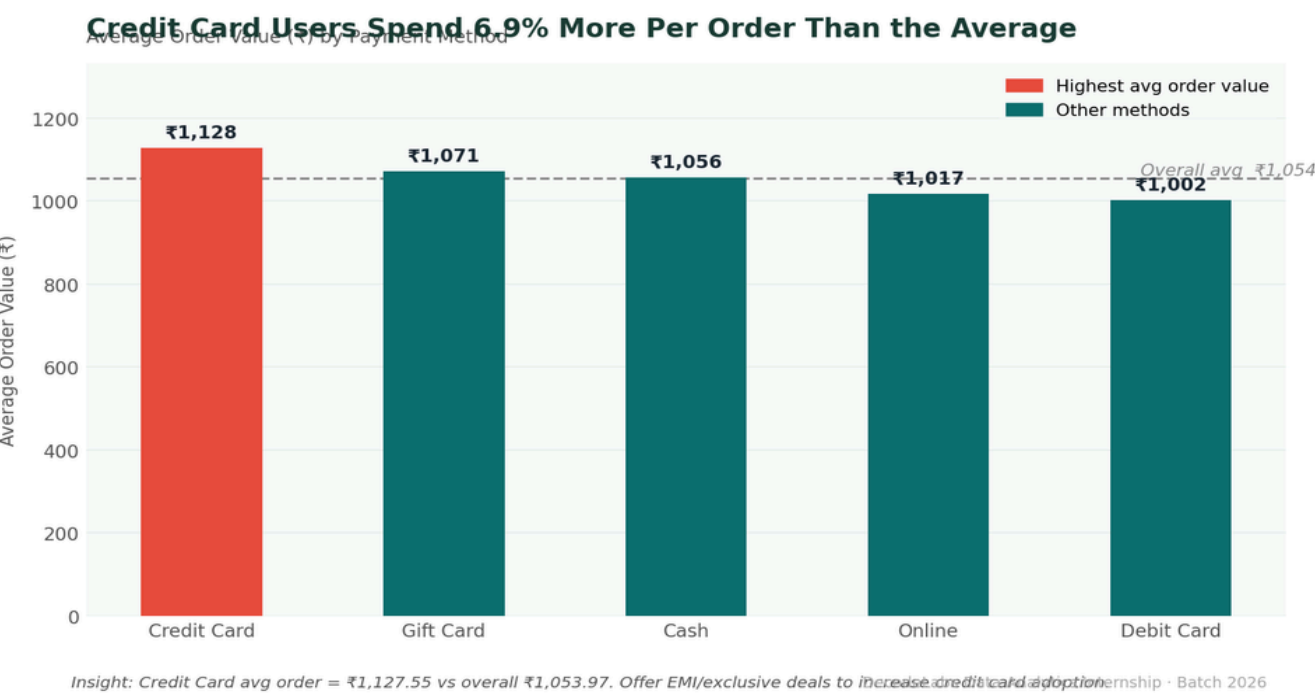
Chart 4 — Revenue by Referral Source



Insight: Instagram outperforms all channels. Increase Instagram ad spend & influencer partnerships to scale this channel further. Batch 2026

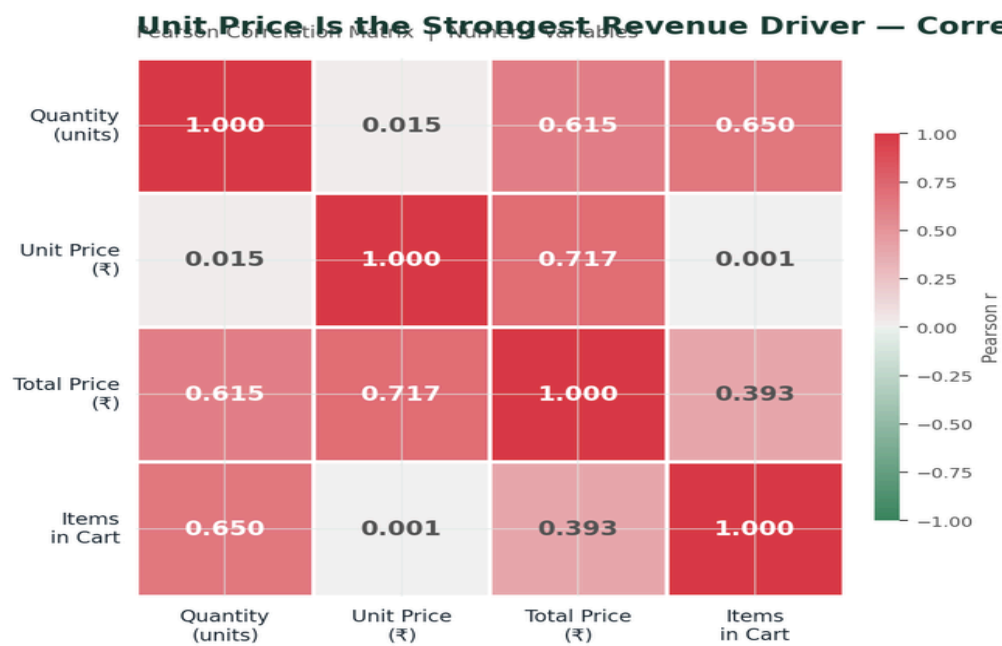
Insight + Recommendation:	Instagram leads with Rs 2,75,285 across 259 orders. Referral programme is the weakest channel at Rs 2,26,816. Recommendation: Scale Instagram budget with influencer campaigns. A/B test referral incentives to improve that channel's conversion.
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Chart 5 — Average Order Value by Payment Method



Insight + Recommendation:	Credit Card users average 1,127.55 per order — 6.9% above the overall mean of 1,053.97. Debit Card is lowest at 1,001.56. Recommendation: Introduce Credit Card exclusive offers and No-Cost EMI options to increase high-value transactions.
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Chart 6 — Pearson Correlation Matrix



Insight: UnitPrice→TotalPrice ($r=0.717$) and Quantity→TotalPrice ($r=0.615$) are key drivers. Premium pricing → higher revenue per order.

Insight + Recommendation: UnitPrice →TotalPrice ($r=0.717$) is the strongest relationship — premium pricing is the #1 revenue lever. Quantity→TotalPrice ($r=0.615$) also strong. Recommendation: Focus on upselling premium SKUs over increasing order volume.

EXECUTIVE SUMMARY — TOP 5 ACTIONABLE FINDINGS

CRITICAL	Fix the 41.4% Revenue Leakage 497 orders Cancelled/Returned. Root cause analysis required immediately. Even a 10% reduction = ~50 recovered orders.
HIGH	Double Down on Instagram Rs 2.75L revenue from 259 orders. Highest volume & value channel. Increase ad spend by 20% and test influencer partnerships.
HIGH	Lead with Premium Products Chair, Printer, Laptop average Rs 1,079-1,111 per order. UnitPrice is the #1 revenue driver ($r=0.717$). Push premium SKUs.
MEDIUM	Capture Mid-Year Revenue Spike May-June consistently peaks every year. Pre-load inventory and launch promotions in April for maximum impact.
MEDIUM	Incentivise Credit Card Payments Credit Card users spend Rs 74 more per order than average. No-Cost EMI and exclusive credit offers will grow basket size.